

ATKIN AND SWINNERTON-DYER CONGRUENCES FOR MEROMORPHIC MODULAR FORMS

Michael Allen, Ling Long, and Hasan Saad

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HECKE'S RECURRENCE RELATIONS

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Consider the q -series

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ANSWER

Such recurrences exist for q -expansions of holomorphic modular forms.

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- ③ Simultaneously diagonalize $S_k(\Gamma_0(N))$.

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SOLUTION

Investigate p -adic recurrence relations.

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If E_4 denotes the Eisenstein series of weight 4, the modular form

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Is this an example of a more general phenomenon?

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Moreover, the roots of $P_{k,p}$ are algebraic integers and have absolute value $p^{\frac{k-1}{2}}$ under any embedding of $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $d = 2 \dim_{\mathbb{C}} S_k(\Gamma)$.

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Moreover, the roots of $P_{k,p}$ are algebraic integers and have absolute value $p^{\frac{k-1}{2}}$ under any embedding of $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $d = 2 \dim_{\mathbb{C}} S_k(\Gamma)$. The congruence relation holds modulo $p^{(k-1)s}$ if we replace $S_k(\Gamma, \mathbb{Z}_q)$ by $S_k^{1-ex}(\Gamma, \mathbb{Z}_q)$.

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where $u_p(C) \in \mathbb{Z}_p^\times$ and

$$u_p(C) + \frac{p}{u_p(C)} = p + 1 - \#C(\mathbb{F}_p).$$

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Consider the Hauptmodul $\lambda(\tau)$ of $\Gamma_1(4)$ defined by $\lambda(\tau) := 16q \cdot \frac{\prod_{n=1}^{\infty} (1+q^{2n})^{16}}{\prod_{n=1}^{\infty} (1+q^n)^8}$ and the usual theta function given by $\theta(\tau) := \sum_{n \in \mathbb{Z}} q^{n^2}$.

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where $\mu_{2,p}$ is the p -adic unit root for the Legendre elliptic curve

$$E_2^{\text{Leg}} : y^2 = x(x-1)(x-2).$$

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In special cases, the longer recurrences reduce to 2-term recurrence relations.

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- ① *We denote by $X = X(\Gamma)$ the algebraic curve $\Gamma \backslash \overline{\mathfrak{H}}$ and by $Y = Y(\Gamma)$ the affine subspace $\Gamma \backslash \mathfrak{H}$. We denote by K an algebraic number field over which X is defined.*

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- ② If $k \geq 3$, $\mathfrak{u} \subset Y(K)$ is finite, and (an appropriate) $M \geq 1$, we denote by $S_k(\Gamma, \star \mathfrak{u}, \mathfrak{o}_K[1/M])$ the space of meromorphic modular forms with poles supported on $\mathfrak{u}$, which vanish at the cusps, and which have Fourier coefficients in $\mathfrak{o}_K[1/M]$.

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GENERIC ASD CONGRUENCES

THEOREM (ALLEN–LONG–SAAD)

Let Γ be a subgroup of finite-index in $\mathrm{SL}_2(\mathbb{Z})$ (with some geometric conditions) such that $X, \mathfrak{u}$ are defined over K . For almost all primes p , there exists

$$P_{k,p^r,\mathfrak{u}}(T) = \sum_{i=0}^d A_{i,p^r,\mathfrak{u}} T^i \in \mathbb{Z}[T]$$

such that for any $f(\tau) = \sum_{n \geq 1} a_f(n) t^n \in S_k(\Gamma, \star \mathfrak{u}, \mathfrak{o}_K \left[\frac{1}{M} \right])$ and any $m, s \geq 1$,

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and where $\gamma_p \in K_p$ depends only on p and Γ .

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where

$$\alpha_u + \beta_u = p^r + 1 - \#E_u(\mathbb{F}_{p^r}), \quad \alpha_u \cdot \beta_u = p^r.$$

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such that for almost all primes p and $m, s \geq 1$ we have

$$p^{k-1} \gamma_p^{mp^s} \sigma(a_{u,i}(mp^{s-1})) \equiv p \cdot c_{u,i} \cdot a_{u,\tilde{\sigma}(i)}(mp^s) \pmod{p^{(k-1)s+j_{f,p}}}$$

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$$d_{u,i}^r + \frac{p^r}{d_{u,i}^r} = p^{ri} \cdot \left(p^{r(k-2-2i)} + 1 - \#E_u(\mathbb{F}_{p^{r(k-2-2i)}}) \right),$$

where

$$d_{u,i} := \sigma^{r-1} \left(c_{u,i}^{\lfloor (r+1)/2 \rfloor} \right) \cdot \sigma^r (c_{u,\tilde{\sigma}(i)}^{\lfloor r/2 \rfloor}).$$

EXAMPLES

EXAMPLE

Let $p \geq 5$ and consider the weight 3 modular forms

$$f(\tau) := \frac{1}{2\pi i} \cdot \frac{\theta(\tau)^2 \cdot \frac{d\lambda(\tau)}{d\tau}}{\lambda(\tau) - 2} =: \sum_{n \geq 1} a_f(n) q^n$$

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where $\mu_p := -\left(\frac{2}{p}\right) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{4})^2$ with Γ_p being the p -adic Gamma function.

EXPLICIT EXAMPLES

THEOREM (ALLEN–LONG–SAAD)

We have

- ① *Explicit method (algebraic formula) to compute all such forms for weights 3 and 4 and group $\Gamma_1(4)$ whose poles are located at CM points of the Legendre elliptic curve.*

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- ② *Pengcheng's Conjectures (analytic constructions) for $\mathrm{SL}_2(\mathbb{Z})$ are true (except possibly at $p = 2$).*
- ③ *The methods of proof of either the analytic or algebraic formulas transfer to all genus 0 finite-index subgroups of $\mathrm{SL}_2(\mathbb{Z})$.*

OVERCOMING LIMITATIONS

PROBLEM

How to overcome analytic (working over $\mathfrak{H}$) limitations to obtain arithmetic information about modular forms?

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SOLUTION

View modular forms as geometric objects.

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where w is the coordinate on $\mathbb{C}/\Lambda$.

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This is a quite geometric picture!

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Given a differential dw on $\mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}\tau)$, as we move from $\tau \rightarrow \gamma\tau$, $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ in $\mathfrak{H}$, dw changes to $(c\tau + d)dw$.

Therefore, a modular form is a “correction term” for $dw^{\otimes k}$ to be defined on $\Gamma \backslash \mathfrak{H}$.

REMARK

This is a quite geometric picture! It also seems to not depend on $\mathbb{C}$.

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Let E be the universal elliptic curve over $Y_1(N) := \Gamma_1(N) \backslash \mathfrak{H}$.

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EXAMPLE

If $N = 4$, then $X_1(N)$ is the projective line $\mathbb{CP}^1$ and the universal elliptic curve is the Legendre family of elliptic curves

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REMARK

The above is not entirely accurate as for $N \leq 5$, we need stacks, but this is essentially the picture.

KODAIRA–SPENCER ISOMORPHISM

QUESTION

In a general course on modular forms, a cusp form of even weight k is explained as a global section

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- ② *Intuitively, ω is a “square-root” of $\Omega_X^1(\log Z)$.*
- ③ *By the above, a **cusp** form of weight k is a global section of $\omega^{k-2} \otimes \Omega_X^1$.*

DE RHAM COHOMOLOGY

DEFINITION

If we have an elliptic curve E/R given by the equation

$$E : y^2 = x^3 + Ax + B$$

the de Rham cohomology $H_{dR}^1(E/R)$ is the R -module defined as

$$R \frac{dx}{y} \oplus R \frac{xdx}{y} \text{ modulo exact forms.}$$

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Analytically, if $E = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ and $R = \mathbb{C}$, then $x = \wp(w)$, $y = \wp'(w)$, and

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the differential forms of the second kind, that is, with poles of order at most 2 at the lattice.

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FACT

There is a sheaf (vector bundle) $\mathcal{E}$ on X whose each fiber (away from the cusps) is the de Rham cohomology of the fiber.

THE GAUSS–MANIN CONNECTION

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Given any derivation D on Y , we have a “derivative”

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EXAMPLE (HYPERGEOMETRIC PICARD–FUCHS DIFFERENTIAL EQUATION)

If $\mathcal{E}$ is the de Rham cohomology of the Legendre elliptic curve, then

$$\nabla\left(\frac{\partial}{\partial\lambda}\right)^2 \frac{dx}{y} = \frac{2\lambda-1}{\lambda(1-\lambda)} \nabla\left(\frac{\partial}{\partial\lambda}\right) \frac{dx}{y} + \frac{1}{4\lambda(1-\lambda)} \frac{dx}{y}.$$

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THEOREM (KATZ)

Locally near the cusp $i\infty$, $\mathcal{E}$ has a basis ω_{can} and $\xi_{can} := \nabla\left(q\frac{d}{dq}\right)\omega_{can}$ such that

$$\nabla\left(q\frac{d}{dq}\right)\xi_{can} = 0.$$

THE “WEAKLY HOLOMORPHIC” CHAIN COMPLEX

QUESTION

If g is a modular form of weight $2 - k$, then $\left(q \frac{d}{dq}\right)^{k-1} g$ is modular of weight k by a direct computation.

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Denote by $\mathcal{E}_{k-2}$ the $(k-2)$ -th symmetric power of $\mathcal{E}$ and by $\Omega^\bullet$ the chain complex

$$\Omega^\bullet = [\mathcal{E}_{k-2} \xrightarrow{\nabla} \nabla(\mathcal{E}_{k-2}) + \mathcal{E}_{k-2} \otimes \Omega_X^1]$$

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The hypercohomology $\mathbb{H}^1(X_1(N), \Omega^\bullet)$ is given by

$$\mathbb{H}^1(X, \Omega^\bullet) \cong \frac{S_k^{!-ex}(\Gamma_1(N))}{\left(q \frac{d}{dq}\right)^{k-1} M_{2-k}^!(\Gamma_1(N))}.$$

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Over $\mathbb{C}$, $S_k^{!-ex} = S_k^!$. The difference is an arithmetic condition when we pass to working over rings.

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Let $\mathfrak{u} \subset Y_1(N)$. Denote by $\Omega_{\mathfrak{u}}^{\bullet}$ the chain complex

$$\Omega_{\mathfrak{u}}^{\bullet} = [\mathcal{E}_{k-2} \xrightarrow{\nabla} \nabla(\mathcal{E}_{k-2}) + \mathcal{E}_{k-2} \otimes \Omega_X^1(\log \mathfrak{u})]$$

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REMARK

We do not really have an isomorphism upstairs. The hypercohomology is a subset of the right-hand side defined by some arithmetic conditions that relate to the pole orders.

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It turns out there is a way to define $\tau \rightarrow p\tau$ acting on our quotient spaces, even though we are no longer dealing with the upper half-plane.

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$$F \left(\sum_{n=-M}^{\infty} a_f(n) q^n \right) = p^{k-1} \cdot \sum_{n=-M}^{\infty} a_f(n) q^{pn} \mod \left(q \frac{d}{dq} \right)^{k-1} \mathbb{Z}_p[[q]].$$

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We have a short exact sequence

$$0 \longrightarrow \mathbb{H}^1(X, \Omega^\bullet) \longrightarrow \mathbb{H}^1(X, \Omega_u^\bullet) \longrightarrow \bigoplus_{u \in \mathfrak{u}} \mathcal{E}_{k-2, u} \longrightarrow 0.$$

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*The Frobenius defined by Scholl extends to $\mathbb{H}^1(X, \Omega_u^\bullet)$. On $\mathcal{E}_{k-2,u}$, it acts as the usual Frobenius on the $(k-2)$ -th symmetric power on the ℓ -adic Tate module of E_u , **twisted by p** .*

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$$P_{k,p,u}(T) = \det(T - F|_{\mathbb{H}^1(X, \Omega^\bullet)}) \cdot \prod_{u \in \mathfrak{u}} \prod_{i=0}^{k-1} (T - p\alpha_u^i \beta_u^{k-2-i}).$$

RESIDUES

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What are the maps in this short exact sequence

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*For the first map, we have **another** identification of $\mathbb{H}^1(X, \Omega^\bullet)$ with modular forms (modulo the Bol operator) with poles supported on $\mathfrak{u}$ and certain conditions on their principal parts.*

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The second map is a residue map computed as follows.

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RESIDUE MAP (BROWN-FONSECA)

The second map is a residue map computed as follows. If $f(\tau)$ is a meromorphic modular form with a pole at $u = (w \bmod \Gamma)$, $w \in \mathfrak{H}$, consider the expression

$$f(\tau) \cdot (X - \tau Y)^k.$$

Take the residue at w and send $X - wY \rightarrow \frac{dx}{y}$ and $Y \rightarrow -\nabla \left(\frac{d}{d\tau}\right) \frac{dx}{y}$.

CM ON MEROMORPHIC FORMS

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STRATEGY

- 1 Find a CM basis for $\mathcal{E}_u = H_{dR}^1(E_u/\mathbb{C})$.

CM ON MEROMORPHIC FORMS

OBSERVATION

If $\dim S_k(\Gamma) = 0$, then $\mathbb{H}^1(X, \Omega^\bullet) = 0$. If $u \in Y(\Gamma)$, this implies that we have an isomorphism

$$\mathbb{H}^1(X, \Omega_u^\bullet) \xrightarrow{\text{Res}} \mathcal{E}_{k-2,u}.$$

Therefore, if E_u is a CM elliptic curve, the space of cusp forms $\mathbb{H}^1(X, \Omega_u^\bullet)$ inherits the CM action.

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- 4 These preimages are eigenvectors of Frobenius.

THE BASIS FOR $SL_2(\mathbb{Z})$

DEFINITION

If $k \in \mathbb{Z}$ and $f(\tau)$ is a function on $\mathfrak{H}$, the raising operator is defined by

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CONJECTURE (ZHANG)

Let $a, b, c \in \mathbb{Z}$ with $\gcd(a, b, c) = 1$, $D = b^2 - 4ac$, $\alpha_D = \frac{-b + \sqrt{-D}}{2a} \in \mathfrak{H}$ such that $j(\alpha_D) \neq 0, 1728$ and consider the elliptic curve

$$E_{j(\alpha_D)} : \quad y^2 + xy = x^3 - \frac{36}{j(\alpha_D) - 1728}x - \frac{1}{j(\alpha_D) - 1728}.$$

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$$f_{k,r,\alpha_D}(\tau) := E_k(\tau) \cdot R_{2-k,z}^r \left(\frac{E_{14-k}(z)}{\Delta(z) \cdot (j(\tau) - j(z))} \right) \Bigg|_{z=\alpha_D} =: \sum_{n \geq 1} a_{k,r,\alpha_D}(n) q^n.$$

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If $p \nmid (k-2)!$ and $m, s \geq 1$, then we have

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- 3 If w denotes the coordinate on $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\alpha_D)$, we can rewrite $\frac{dx}{y} = dw$ and $\frac{x dx}{y} = \wp_{\alpha_D}(w)dw$, where $\wp$ is the Weierstrass $\wp$ -function.

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- 4 For the form $\eta = \text{Res}(f_{k,r,\alpha_D})$, show that $\eta - [\alpha_D]^*\eta$ has vanishing periods.
- 5 A vanishing period with zero residues implies exactness and thus vanishing in the de Rham cohomology.

GENERIC ASD CONGRUENCES

THEOREM (ALLEN–LONG–SAAD)

Let Γ be a subgroup of finite-index in $\mathrm{SL}_2(\mathbb{Z})$ (with some geometric conditions) such that X is defined over K . For almost all primes p , there exists a polynomial

$$P_{k,p^r,u}(T) = \sum_{i=0}^d A_{i,p^r,u} T^i \in \mathbb{Z}[T]$$

such that for any $f(\tau) = \sum_{n \geq 1} a_f(n) t^n \in S_k(\Gamma, \star u, \mathfrak{o}_K \left[\frac{1}{M} \right])$ and any $m \geq 1, s \geq 1$, we have

$$\sum_{i=0}^d p^{r(k-1)i} \cdot A_{i,p^r,u} \cdot \gamma_p^{mp^s(p^{ri}-1)/(p-1)} \cdot a_f \left(\frac{mp^s}{p^{ri}} \right) \equiv 0 \pmod{p^{r(k-1)s+j_{f,p}}},$$

where

$$j_{f,p} := \begin{cases} k + \mathrm{ord}_u(f) & \text{if } \dim S_k(\Gamma) = 0 \text{ and } -\mathrm{ord}_u(f) \leq k-1, \\ -\mathrm{ord}_p((- \mathrm{ord}_u(f) - 1)!) & \text{otherwise} \end{cases}$$

and where $\gamma_p \in K_p$ depends only on p and Γ .

SPECIAL ASD CONGRUENCES

COROLLARY

Suppose $\dim S_k(\Gamma) = 0$ and E_u is a CM elliptic curve by an imaginary quadratic subfield of K . There exist modular forms

$$\left\{ f_{u,i} = \sum_{n \geq 1} a_{u,i}(n)t^n \in S_k(X, \star u, \mathfrak{o}_K \left[\frac{1}{M}, u, \frac{1}{u} \right]) \right\}$$

such that for almost all primes p and $m, s \geq 1$ we have

$$p^{k-1} \gamma_p^{mp^s} \sigma(a_{u,i}(mp^{s-1})) \equiv p \cdot c_{u,i} \cdot a_{u,\tilde{\sigma}(i)}(mp^s) \pmod{p^{(k-1)s+jf,p}}$$

where σ is the Frobenius endomorphism of K_p , $c_{u,i} \in p^{k-2-i} \mathfrak{o}_{K_p}$, and $\tilde{\sigma}(i) := i$ if σ fixes the CM field in K_p and $\tilde{\sigma}(i) := k-2-i$ otherwise. If $0 \leq i \leq \frac{k-2}{2}$, then we have that

$$d_{u,i}^r + \frac{p^r}{d_{u,i}^r} = p^{ri} \cdot \left(p^{r(k-2-2i)} + 1 - \#E(\mathbb{F}_{p^{r(k-2-2i)}}) \right),$$

where

$$d_{u,i} := \sigma^{r-1} \left(c_{u,i}^{\lfloor (r+1)/2 \rfloor} \right) \cdot \sigma^r (c_{u,\tilde{\sigma}(i)}^{\lfloor r/2 \rfloor}).$$

EXPLICIT RESULTS

THEOREM

- 1 *Explicit eigenforms of weights 3 and 4 constructed algebraically for $\Gamma_1(4)$.*
- 2 *Explicit eigenforms of weights 4, 6, 8, 10, 14 constructed analytically for $\mathrm{SL}_2(\mathbb{Z})$.*